Digital Audio Synthesizer

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Abstract

This project implements a digital synthesizer that converts MIDI note events into playable audio and applies a configurable sequence of signal-processing effects. The synthesis subsystem combines sine, square, triangle and sawtooth oscillators, detuned voices, an attack-hold-decay-sustain-release (AHDSR) amplitude envelope and a resonant voltage-controlled filter (VCF). The MIDI renderer interprets tempo-aware note events and places overlapping notes on a common sample timeline. The effects subsystem provides a class-based rack supporting three saturation algorithms, dynamic-range compression, delay, phaser, flanger, chorus, algorithmic and convolution reverb, VCF, multiband vocoding and pitch shifting; modulated variants of the relevant delay and phase effects are included. The implementation was exercised using the supplied MIDI and audio assets, and the results are reported using saved audio files, waveforms, spectra and amplitude statistics.

1. System overview

The signal path consists of MIDI event interpretation, per-note frequency generation, amplitude-envelope shaping, temporal summation of simultaneously active voices, optional oscillator filtering, and a user-configurable serial effects rack. Independently, the effects can operate on the supplied musical audio recording. All audio processing and measurements use numeric samples rather than live audio-device output.

Table 1. Input and output of the main processing modules.

Module	Input	Output
Synthesis	MIDI pitches, timings and velocities	Mono audio at 44,100 samples/s
Wave shaping	Waveform type, frequency, phase, detune	Periodic tone and AHDSR-shaped voice
Effects rack	Audio plus effect parameters	Serially processed audio with wet/dry control
Analysis	Waveforms, effect outputs, WAV recordings	Spectra, RMS, peak and comparison plots

2. Experimental configuration

The synthesis and processing samples are generated and stored at 44.1 kHz. The source for independent effects is the supplied Ave-Maria.wav, an eight-second mono musical signal. The MIDI demonstration uses newSong.mid. The four room impulse responses are read from Room1.wav through Room4.wav; resampling and mono conversion are applied when required. The multiband vocoder processes the complete 48.04-second Carrier.mp3 and Modulator.mp3 source pair. The final saved WAV files use PCM-16 format; floating-point signals are constrained to a common output peak limit before encoding.

Table 2. Experimental configuration and source recordings.

Quantity	Value / design choice
Sampling rate	44,100 Hz
Base music-tuning reference	A4 = 440 Hz; 12-tone equal temperament
MIDI rendering	11 extracted note events, one channel; two synthesizer timbres
Synthesis output	4.17 s including the release tail
Independent musical effects	8.00 s source audio
Vocoder example	48.04 s complete carrier/modulator recordings
Convolution reverb	Four supplied room impulse-response recordings

3. Oscillator and detuning design

The sinusoidal oscillator calculates x[n] = A sin(2 pi f n / fs + phi), where fs is the sampling rate. The other waveform shapes use periodic square, symmetric triangle and sawtooth functions at the same phase rate. Sine has a single spectral component at the fundamental, whereas the other waveforms contain harmonics; the square waveform emphasizes odd harmonics, the triangular waveform has attenuated odd harmonics, and the sawtooth waveform contains both even and odd harmonics. Digital discontinuous waveforms exhibit spectral components approaching Nyquist and may alias at higher fundamental frequencies; the plotted outputs characterize the implemented discrete-time waveforms rather than an ideal band-limited oscillator.

A detuned voice is formed as a weighted sum of oscillators with nearby frequencies. For a detuning offset c measured in cents, f(c) = f0 2^(c/1200). The demonstration combines -9, 0 and +9 cents with relative weights 0.75, 1.00 and 0.75. The voice amplitude is normalized by the sum of the absolute weights. The same mechanism is used in the main MIDI rendering, with -7, 0 and +7 cents. An alternative MIDI loader and synthesizer class render the complete melody again using triangle waves with -15, 0 and +15 cent offsets.

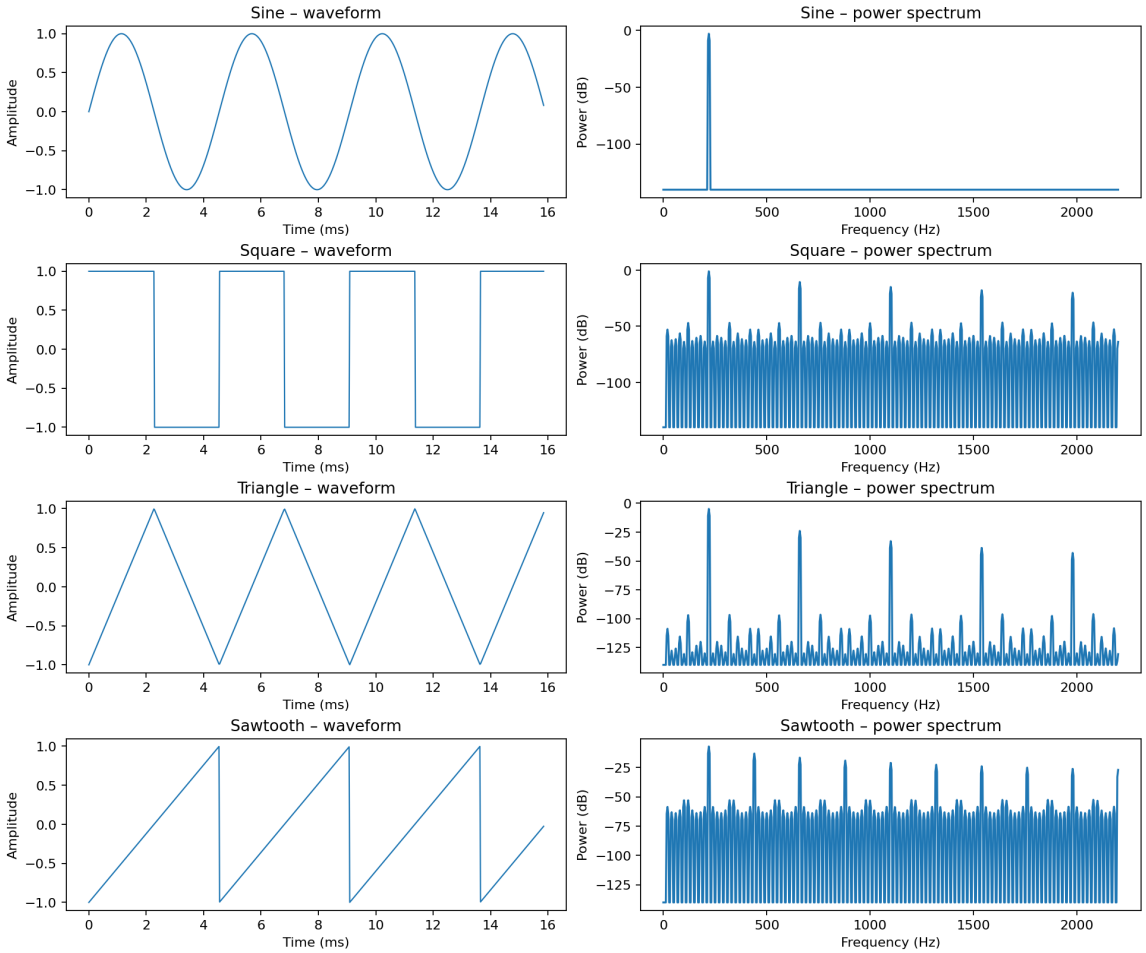


Figure 1. Time-domain waveforms and periodogram-derived spectra for the four oscillators.

4. AHDSR amplitude envelope

The envelope defines five stages for every note: an attack from zero to full level, a short hold, a decay to the sustain level, a constant sustain until the note-off event, and a release to zero. Each stage is calculated in sample units so its duration is controlled precisely by the sampling frequency. Short notes are handled by truncating attack/hold/decay while preserving a release from the actual final note-on level. Multiplying each oscillator output by its own envelope reduces discontinuities between note boundaries and controls perceptual articulation.

Table 3. Amplitude-envelope settings in the MIDI renderer.

Envelope parameter	MIDI-rendering value
Attack	8 ms
Hold	12 ms
Decay	100 ms
Sustain level	0.68 (linear gain)
Release	120 ms after note-off

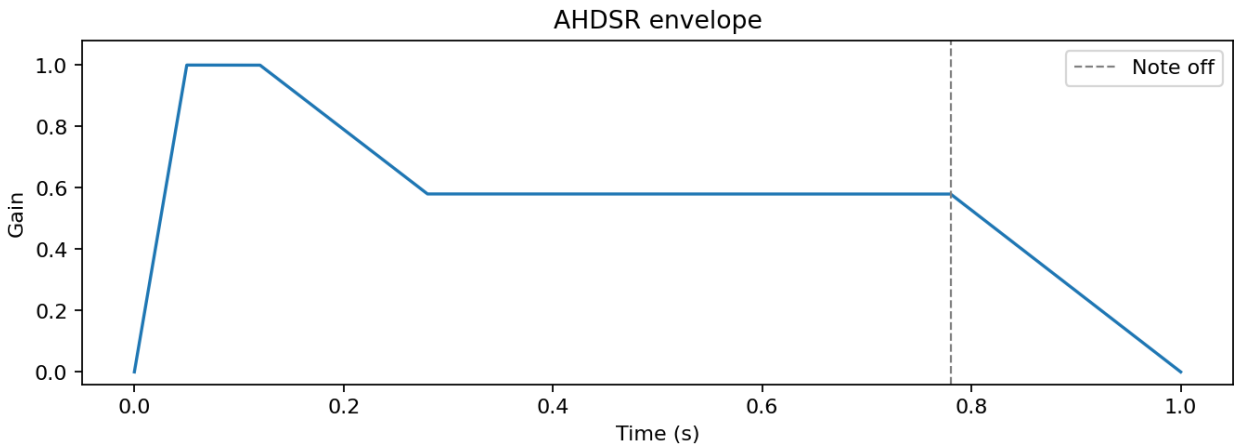


Figure 2. Example AHDSR trajectory with a clearly marked note-off and release interval.

5. MIDI decoding and polyphonic synthesis

A MIDI message stream stores event changes rather than recorded audio. The parser obtains absolute event times from the merged tempo-aware event sequence. Notes are keyed by MIDI channel and pitch; a positive-velocity note-on opens an active note, and a note-off (or note-on with zero velocity) closes it. Matching events produce note start time, end time, channel, pitch and velocity. Pitch numbers are mapped to physical frequency using f(n) = 440 * 2^((n - 69)/12), where n is the MIDI note number.

Each event is rendered into its own detuned and enveloped waveform, scaled by velocity/127 and placed at the corresponding sample offset in a shared output buffer. The overlap of events is mixed by sample-wise addition. A release tail is retained beyond the final MIDI note-off. This approach preserves real event timing and allows simultaneous notes, unlike note concatenation.

Table 4. MIDI pitches and note timings extracted from newSong.mid.

MIDI note	Frequency (Hz)	Start (s)	End (s)
43	98.00	0.00	0.50
50	146.83	0.50	1.00
50	146.83	1.00	1.50
46	116.54	1.50	2.00
45	110.00	2.00	2.50
45	110.00	2.50	2.75
43	98.00	2.75	3.00
45	110.00	3.00	3.25
43	98.00	3.25	3.50
45	110.00	3.50	3.75
46	116.54	3.75	4.00

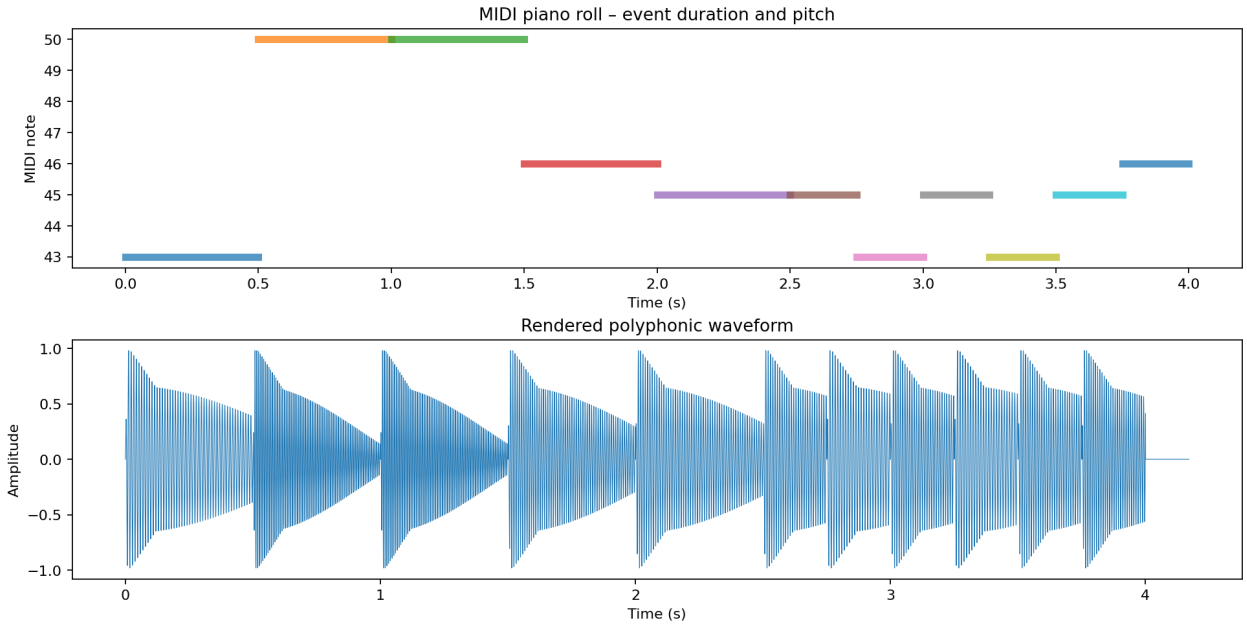


Figure 3. Extracted MIDI note durations and the synthesized audio waveform. The waveform extends beyond the final note-off due to release shaping.

6. Resonant low-pass filtering

The VCF block applies a second-order low-pass section parameterized by cutoff frequency and Q (resonance). For an angular cutoff omega = 2 pi fc/fs and alpha = sin(omega)/(2Q), a normalized biquad implementation uses coefficients derived from (1 - cos(omega))/2, 1 - cos(omega), (1 - cos(omega))/2 in the numerator and 1 + alpha, -2 cos(omega), 1 - alpha in the denominator. Output samples are calculated using the causal recursive filter. The 650 Hz oscillator-filtering demonstration illustrates the smoothing of abrupt square- and sawtooth-wave transitions and the associated reduction of high-frequency energy.

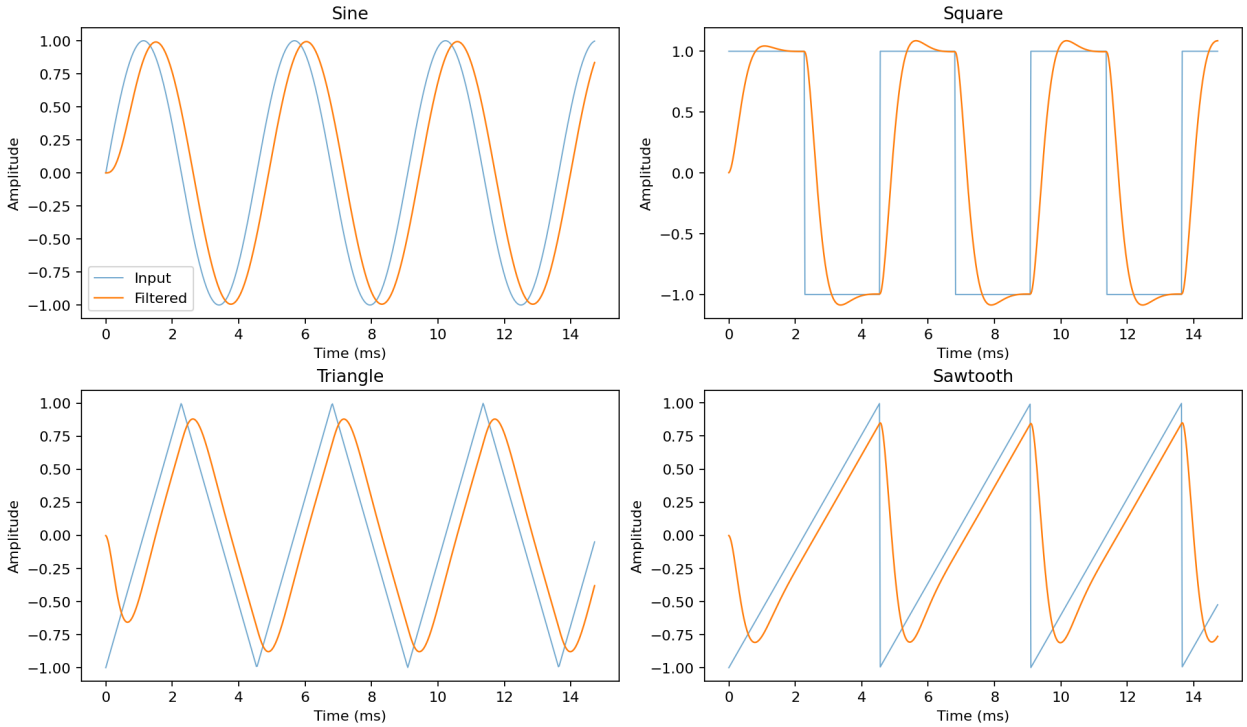


Figure 4. Each oscillator before and after low-pass filtering at a 650 Hz cutoff.

7. Software architecture of the effects rack

The effect architecture consists of an Effect base class and an EffectRack container. Each subclass defines a signal transformation, while the base class handles per-instance input gain, output gain, wet/dry blending and bypass. If s is the input, u the transformed input, g_in the input gain, g_out the output gain and w the wet mix, the general structure is y = g_out [w F(g_in s) + (1 - w) s]. The rack processes effects in the order they are supplied; order therefore influences the final result. For convolution reverb, the dry path is zero-padded to match the explicitly retained reverberation tail.

The supported classes are HardClipping, SoftClipping, VariableSaturation, Compressor, Delay, Delay_with_lfo, Phaser, Phaser_with_lfo, Flanger, Flanger_with_lfo, Chorus, Chorus_with_lfo, Reverberation, ConvolutionReverb, VoltageControlledFilter, Vocoder, PitchShifter, and the EffectRack container.

8. Nonlinear dynamics and echo effects

8.1 Hard, soft and variable saturation

Hard clipping limits amplitudes to a symmetrical threshold; here, the threshold is 0.30 with input gain 1.6 and a 0.95 wet fraction. It directly truncates the original waveform above the threshold and creates additional harmonic content. Soft clipping uses a smooth hyperbolic-tangent transfer characteristic, y = T tanh(x/T), with T = 0.33 and input gain 1.8. Variable saturation applies drive, a small DC bias and a shaped tanh nonlinearity, allowing progressively increased harmonic distortion without an abrupt clipping boundary. The three methods are stored as separate WAV outputs and are not merged into one approximation.

8.2 Dynamic-range compression

The compressor estimates an amplitude envelope from the absolute-value signal, with different attack and release smoothing coefficients. Envelope level is converted to decibels. Above the -21 dBFS threshold, a 4:1 ratio attenuates the excess level; a time-varying linear gain is then applied to the source. The demonstration uses 10 ms attack and 120 ms release. Since an automatic makeup gain is not enabled, the output RMS level is lower than the input RMS level, as expected for the configured threshold and ratio.

8.3 Fixed delay and LFO-modulated delay

The ordinary delay reads an older signal sample after a fixed 210 ms period and includes feedback 0.45. The modulated form changes the read offset using a sinusoidal LFO with a center delay of 190 ms, a modulation depth of 12 ms and a 0.35 Hz rate. Fractional-delay linear interpolation avoids rounding the continuously changing read pointer to a single integer sample. A low-pass post-filter with a 2.2 kHz cutoff is applied to the modulated delay path. The examples distinguish fixed repeat spacing from slowly varying timing and spectrum.

9. Phase and time-modulation effects

9.1 Phaser and LFO phaser

A phaser obtains frequency-dependent phase displacement through cascaded first-order all-pass sections. Each section follows a[n] = (tan(pi fc/fs) - 1)/(tan(pi fc/fs) + 1), with the section center frequency fc set to a fixed value or modulated by an LFO. Dry/wet cancellation creates spectral notches whose locations depend on the all-pass phase shift. The LFO implementation uses a 0.6 Hz rate, modulation depth 0.6, four all-pass sections and a modest feedback term of 0.18. A fixed-coefficient version is also exported for comparison.

9.2 Flanger and LFO flanger

The flanger uses a short feedback delay with a nominal length of 6 ms, variation depth of 1.6 ms and feedback coefficient 0.32. Its fixed-delay demonstration retains the short comb-filter structure. The modulated variant changes the read position at 0.3 Hz, producing a moving comb spectrum. Both versions use fractional-delay interpolation and maintain the same number of output samples as the input.

9.3 Chorus and LFO chorus

The chorus uses a delayed copy of the input with a longer time offset: approximately 28 ms and 4 ms modulation depth. The fixed-delay path acts as a simple doubling reference. The modulated implementation averages two delay voices with slightly different center delays, modulation depths, rates and phase, producing a richer composite signal. The examples preserve the separate fixed and modulated implementations.

10. Reverberation techniques

10.1 Algorithmic digital reverb

The algorithmic reverb forms a short reverberant response from four feedback comb-delay paths, with delay times 29, 37, 41 and 53 ms and a feedback coefficient of 0.52. The delayed paths are averaged and mixed with the original signal using a configurable wet proportion. The configuration provides a simple synthetic decay and decorrelates temporal copies without relying on an external room measurement.

10.2 Convolution reverberation

The convolution approach uses the supplied room impulse-response recordings Room1 through Room4. If h[n] is a room response and x[n] the musical input, the wet output is the discrete convolution y[n] = sum_k x[k] h[n-k]. Each response is resampled to 44.1 kHz when necessary, reduced to mono by channel averaging and amplitude-scaled before convolution. FFT convolution efficiently produces the full temporal response. The reverb tail increases duration beyond the eight-second source; the four output lengths range from 9.30 to 11.28 seconds.

11. Spectral processing and pitch transformation

11.1 Voltage-controlled low-pass effect

The low-pass effect is also available as a rack component, using a 950 Hz cutoff and Q = 0.88 in its standalone demonstration. Applied to musical source audio, the filter reduces high-frequency spectral density while retaining a significant portion of the lower-frequency content. This measurement is distinct from the 650 Hz filtering of individual oscillator waveforms.

11.2 Multiband vocoder

The vocoder implements fourteen logarithmically spaced analysis bands between approximately 100 Hz and 8 kHz. Carrier and modulator audio are filtered separately in each band with third-order Butterworth band-pass sections. The magnitude envelope of the modulator band is estimated using a Hilbert analytic signal and low-pass smoothing with a 14 ms time constant. The carrier band is amplitude-modulated by this smoothed envelope and summed across all bands. The output is RMS-scaled to the selected nominal level. Both supplied source recordings are processed at their complete 48.04-second duration.

11.3 Pitch shifting

A pitch shifting effect is implemented with a Short-Time Fourier Transform and phase-vocoder processing followed by a resampling step. The example raises the original source by four semitones while approximately preserving its eight-second duration. Phase-consistent spectral reconstruction is important in this method because directly changing playback speed would also change duration.

12. Audio measurements and results

Table 5 summarizes the saved effects on the eight-second source using duration, normalized sample peak, root-mean-square (RMS) amplitude and RMS level in dBFS. RMS is computed as sqrt(mean(x[n]^2)) and level as 20 log10(RMS). The metrics describe measurable changes in output amplitude; they do not serve as perceptual quality scores. A common output peak ceiling prevents PCM clipping when the signal is saved.

Table 5. Amplitude and duration measurements for the source and main effects.

Recording	Seconds	Peak	RMS	RMS (dBFS)
Source	8.00	0.980	0.318	-9.95
Hard Clipping	8.00	0.334	0.228	-12.85
Soft Clipping	8.00	0.362	0.233	-12.66
Variable Saturation	8.00	0.980	0.418	-7.57
Compressor	8.00	0.980	0.101	-19.91
Delay	8.00	0.882	0.212	-13.49
Delay Lfo	8.00	0.855	0.231	-12.72
Phaser	8.00	0.980	0.281	-11.03
Phaser Lfo	8.00	0.980	0.183	-14.77
Flanger	8.00	0.980	0.245	-12.22
Flanger Lfo	8.00	0.980	0.200	-13.98
Chorus	8.00	0.932	0.250	-12.03
Chorus Lfo	8.00	0.927	0.191	-14.37
Digital Reverb	8.00	0.599	0.144	-16.84
Vcf	8.00	0.980	0.316	-10.01
Pitch Shift	8.00	0.980	0.188	-14.53
Full Rack	8.00	0.341	0.071	-22.99

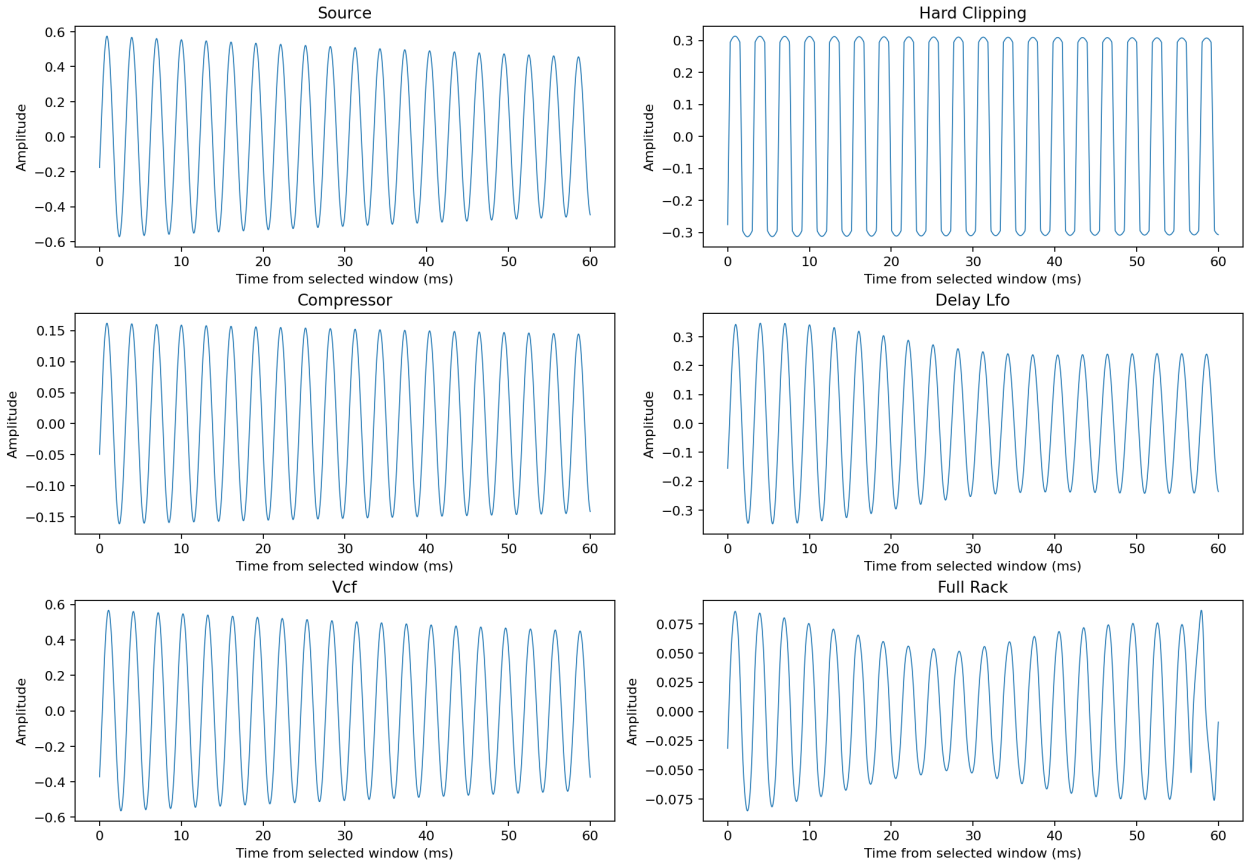


Figure 5. Short time-domain extracts from the source, selected processing paths and complete rack output.

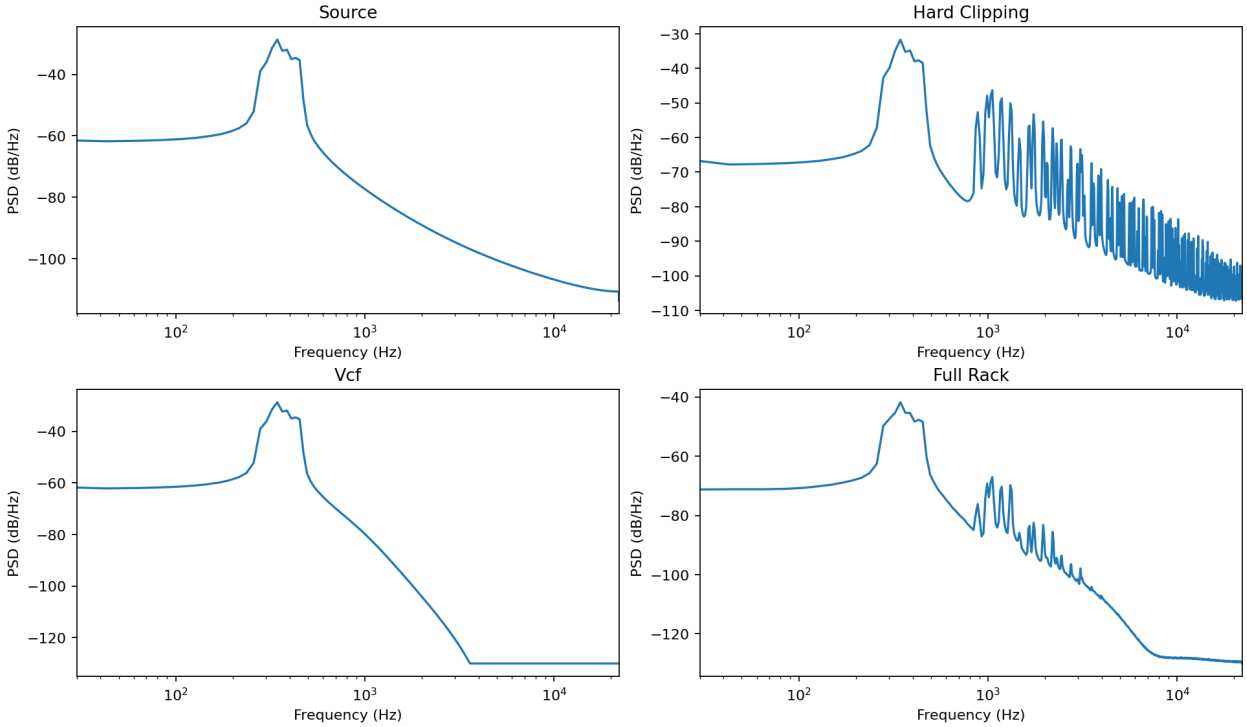


Figure 6. Welch power spectral density estimates illustrate additional clipping harmonics and VCF suppression of high-frequency energy.

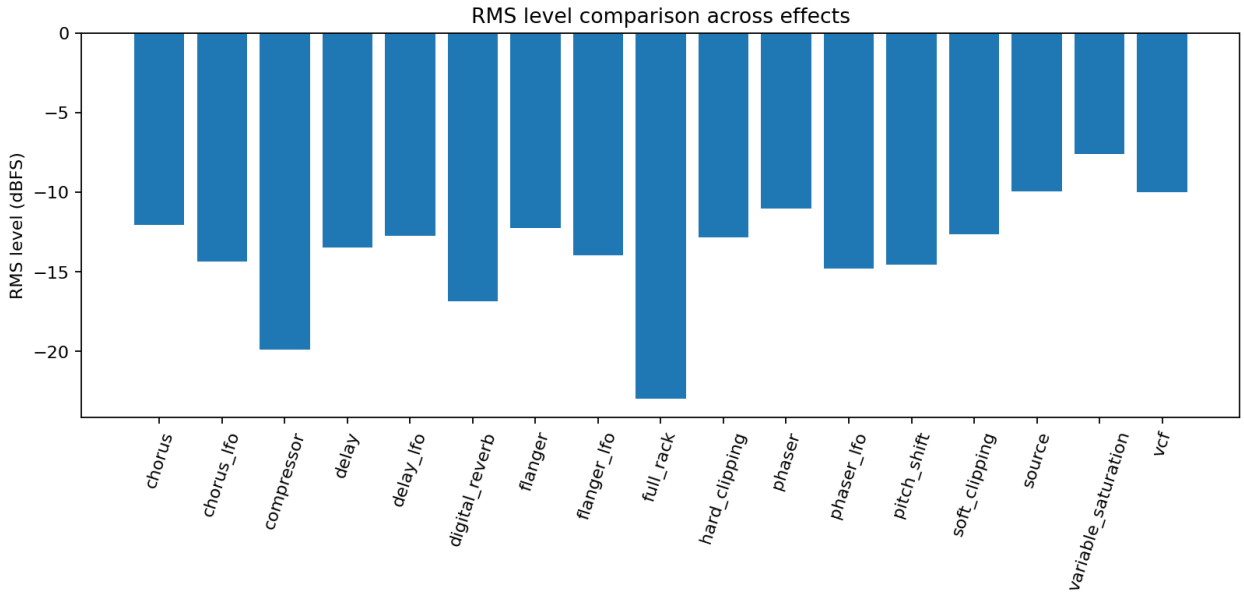


Figure 7. Measured RMS levels for the original musical recording and the core effect demonstrations.

12.1 Spectral changes introduced by processing

A frequency-domain comparison was performed on the exported eight-second signals using Welch power spectral densities (Hann window, 8192-sample segments, 50% overlap). For each signal, one-sided power spectral density was integrated from 2 kHz to the Nyquist frequency. The table reports 10 log10(P_effect/P_source), where zero denotes unchanged high-frequency-band power in the saved recording. Since the input and processed recordings share the same duration and sample rate, these measurements represent the combined effects of processing parameters, wet/dry mixing and any final peak scaling.

Table 6. Measured changes in spectral power above 2 kHz.

Saved recording	Band-power change vs. source (dB)
Hard clipping	+22.39
Soft clipping	+19.74
Variable saturation	+6.35
Compressor	-4.39
VCF, cutoff 950 Hz	-21.21
Full effects rack	-6.16

Hard and soft clipping increase high-frequency-band power through nonlinear harmonic generation. The 950 Hz low-pass filter yields the largest reduction in this band, consistent with the spectral comparisons in Figure 6. The compressor and complete effects rack also reduce this measured band power, although their results combine signal transformation with the selected wet/dry and gain settings. These values quantify the exported signals and should not be interpreted as perceptual audio-quality scores.

13. Room-response, vocoder and synthesis outputs

The impulse-response convolution examples retain their extended wet-signal tails. The vocoder differs in length from the eight-second music demonstrations because it uses the full carrier/modulator recording pair. The MIDI-synthesized output is generated from its own note-event data and is therefore treated as an independent result.

Table 7. Convolution, vocoder and synthesized-audio measurements.

Recording	Seconds	Peak	RMS	RMS (dBFS)
Convolution Room1	10.39	0.684	0.145	-16.76
Convolution Room2	9.30	0.980	0.165	-15.66
Convolution Room3	11.28	0.980	0.173	-15.25
Convolution Room4	9.31	0.980	0.189	-14.46
Vocoder	48.04	0.980	0.111	-19.07
Midi Synth	4.17	0.980	0.443	-7.07
Midi Alternate	4.17	0.979	0.303	-10.36
Osc Detuned	1.00	0.980	0.421	-7.51
Ahdsr Note	0.75	0.980	0.362	-8.83

The individual oscillator outputs and VCF-processed variants were additionally saved for sine, square, triangle and sawtooth waveform families. These demonstrations make it possible to listen directly to harmonic differences and to the effect of low-pass filtering on each wave shape.

14. Multistage processing experiment

The final rack example connects soft clipping (wet = 0.58), 3:1 compression at -17 dBFS (wet = 0.80), a 4.2 kHz resonant low-pass stage (wet = 0.70), LFO chorus (wet = 0.25), a 190 ms feedback delay (wet = 0.26), and algorithmic reverberation (wet = 0.20). The order is significant: saturation creates or strengthens harmonics before VCF filtering, compression changes the amplitude envelope before time-delay effects, and reverberation is the final stage. The combined recording has an RMS amplitude of 0.0709 (-22.99 dBFS) compared with 0.3181 (-9.95 dBFS) for the input; that difference reflects the actual selected gains and wet/dry settings.

15. Validation and conclusion

The notebook checks MIDI tuning at A4 and A5 (440 and 880 Hz), amplitude-envelope bounds and release endpoints, valid note-event timing, effect bypass behavior, finite sample values and successful export of audio and measurement files. The saved outputs comprise 34 WAV recordings, including oscillator and envelope demonstrations, two MIDI renderings, the individual effects, four convolution responses, the full-length vocoder and the cascaded effects rack.

The results establish a reproducible MIDI-to-audio synthesizer with independently configurable signal-processing effects. The generated waveforms, measured spectra and audio-level tables document the behavior of the implementations under the settings reported above. The notebook reproduces these results from the included MIDI and audio assets at a 44.1 kHz sampling rate.